



CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

Lecture No.- 08

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Recap of Previous Lecture



Topic

Logic Part-07



Topics to be Covered



Topic

Logic Part-08



Nested quantifier:

→ Tool which is used to find the truth value in quantity.

$$D: \{1, 2, 3\}$$

$$\forall x (x^2 + y^2 = 3)$$

$$(1^2 + y^2 = 3) \text{ o.s.}$$

$$D: \{1, 2, 3\}$$

$$P(x, y): x \times y \leq 9$$

$$D: \{1, 2, 3\}$$

$$\forall x \forall y (x \times y \leq 9)$$

$$\underline{x=1} \quad \underline{y=1}$$

$$\underline{1 \times 1 \leq 9 (T)}$$

$$\underline{x=1} \quad \underline{y=2}$$

$$\underline{1 \times 2 \leq 9}$$

$$\underline{x=1} \quad \underline{y=3}$$

$$1 \times 3 \leq 9$$

$$x=2 \quad y=1$$

$$2 \times 1 \leq 9$$

$$\forall x \forall y. P(x, y)$$

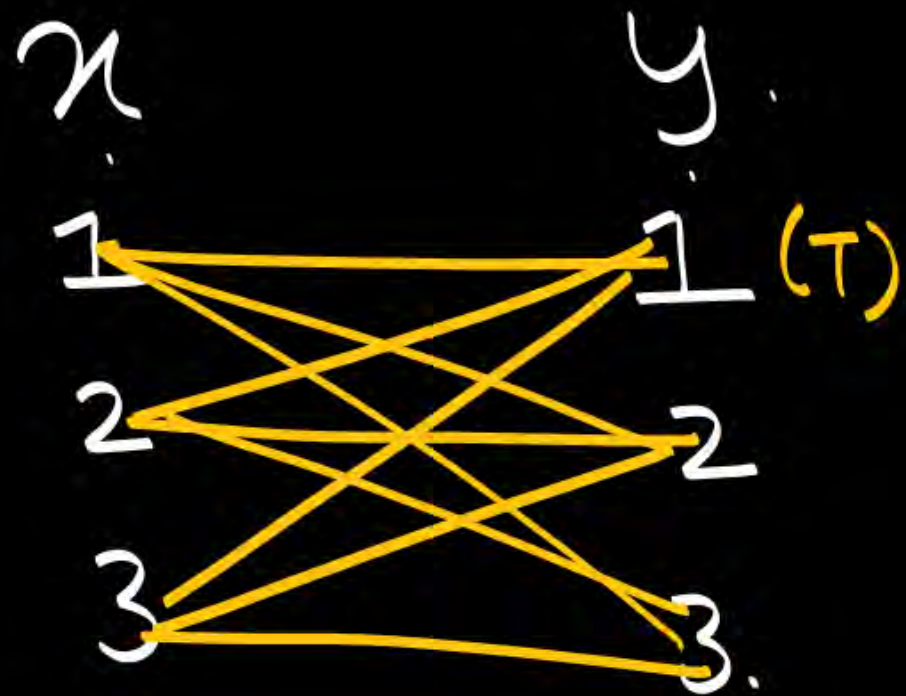
all values of x , all values of y ,
such that $P(x, y)$

for each value of x , each value of
 y , such that
 $P(x, y)$.

for $x = 1$ to 3.
for $y = 1$ to 3.
 $x \times y \leq 9$

$D: \{1, 2, 3\}$

$\forall x \forall y (x \times y \leq 9)$



$\forall x \forall y \rightarrow \text{True}$

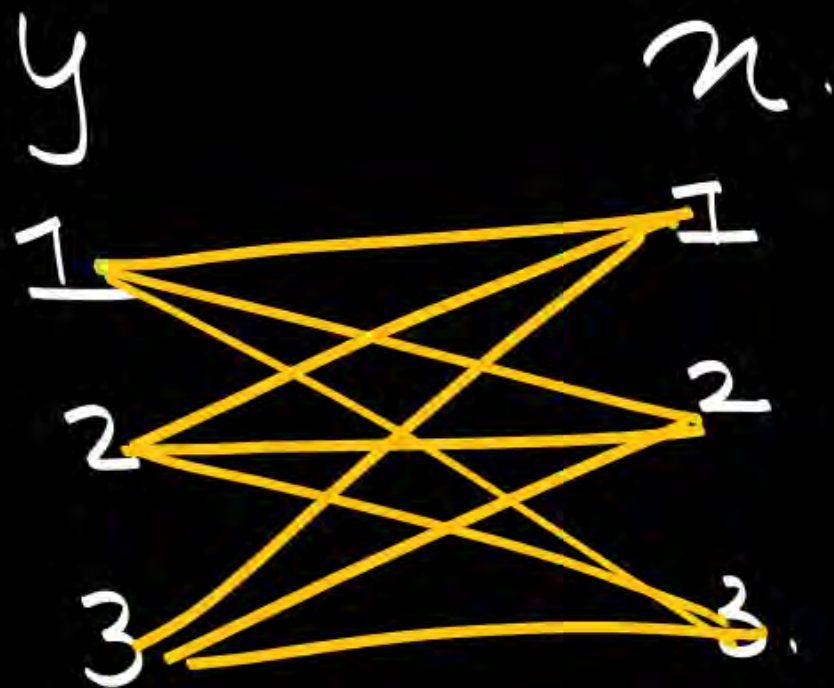
when all edges are True
(all pairs)

when for at least 1 edge
when it is false

$\forall x \forall y (\text{false})$

$D: \{1, 2, 3\}$

$\forall y \forall x (x \times y \leq 9)$



$$\forall x \forall y \equiv \forall y \forall x$$

$$D: \{1, 2, 3\}$$

$$x \times y = 4$$

$$\exists x \exists y (x \times y = 4)$$

$$x=1 \quad y=1$$

$$1 \times 1 = 4 (F)$$

$$x=1 \quad y=3$$

$$1 \times 3 = 4 (F)$$

$$x=1 \quad y=2$$

$$1 \times 2 = 4 (F)$$

$$x=2 \quad y=1$$

$$2 \times 1 = 4 (F)$$

$$\exists x \exists y (x \times y = 4)$$

there exist x , there exist y .

for which $P(x, y)$ is True

at least 1 value of x , at least 1

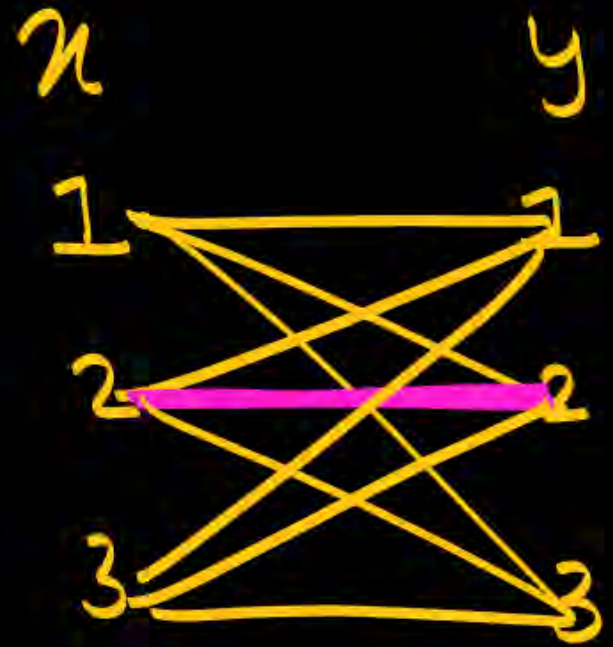
value of y such that $P(x, y)$.

$$x=2 \quad y=2$$

$$2 \times 2 = 4 (T)$$

$$D: \{1, 2, 3\}$$

$$\exists x \exists y (x \times y = 4)$$



$$D: \{1, 2, 3\}$$

$$\exists y \exists x (x \times y = 4)$$

y	x
1	1
2	2
3	3

$\exists x \exists y \rightarrow \text{True}$
when it is True for
at least 1 edge.

$\exists x \exists y \rightarrow \text{false}$
when all edges are false.

$$\exists x \exists y \equiv \exists y \exists x$$

$$\forall x \forall y \equiv \forall y \forall x.$$

$$\exists x \exists y \equiv \exists y \exists x.$$

D: $P(x, y)$:

$$\underline{\forall x \forall y} \rightarrow \underline{\exists x \exists y}:$$

$$\forall x \forall y \rightarrow \exists y \exists x.$$

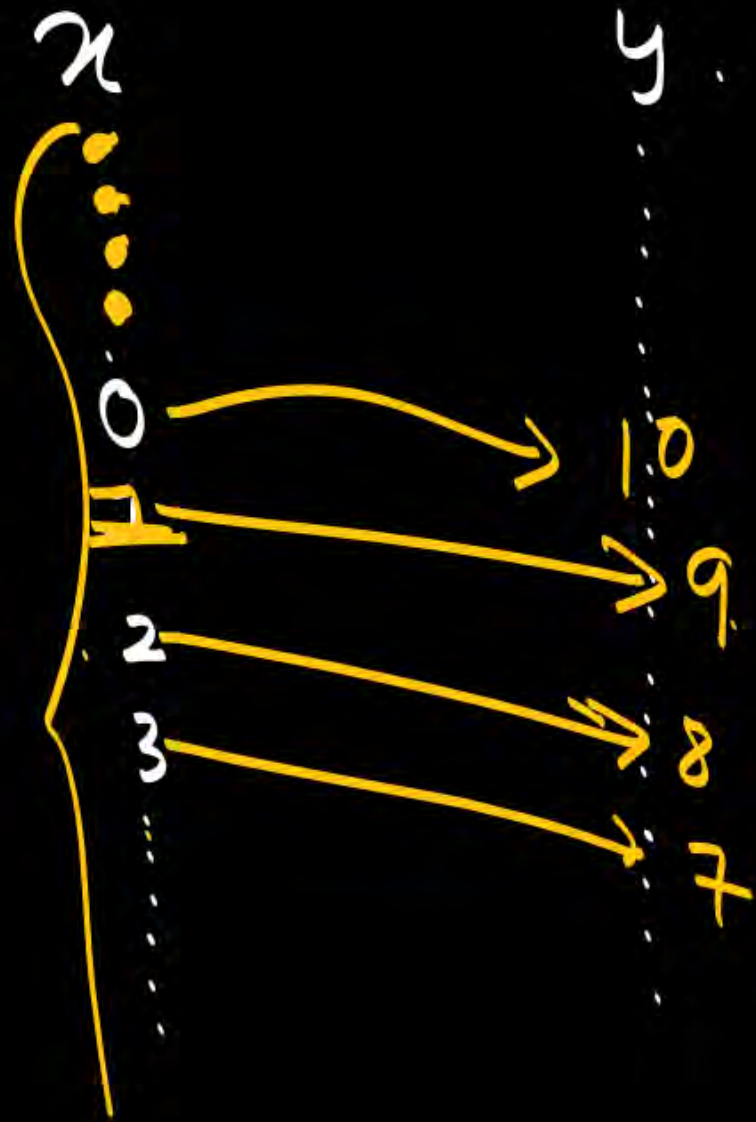
$$\forall y \forall x \rightarrow \exists x \exists y.$$

$$\underline{\forall y \forall x} \rightarrow \underline{\exists y \exists x}$$

True

$D: \mathbb{Z}$

$\forall x \exists y (x + y = 10)$ $\rightarrow$ True



$x = 1$
 $x + y = 10$
 $1 + y = 10$
 $y = 9$

$\forall x \exists y$

for all value of x , there exist y .

for every values of x , some values of y .

D: $\mathbb{Z}$.

$$\exists y \forall x (x + y = 10) \quad \text{false.}$$

$\exists y \forall x$.
there exist y
all values of x .

$$\forall x \exists y (x + y = 10)$$

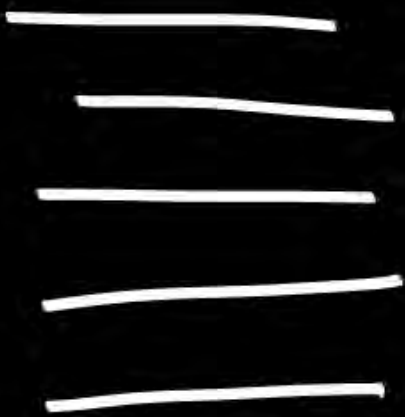
y x .

$$\exists y \forall x (x + y = 10)$$

Reality

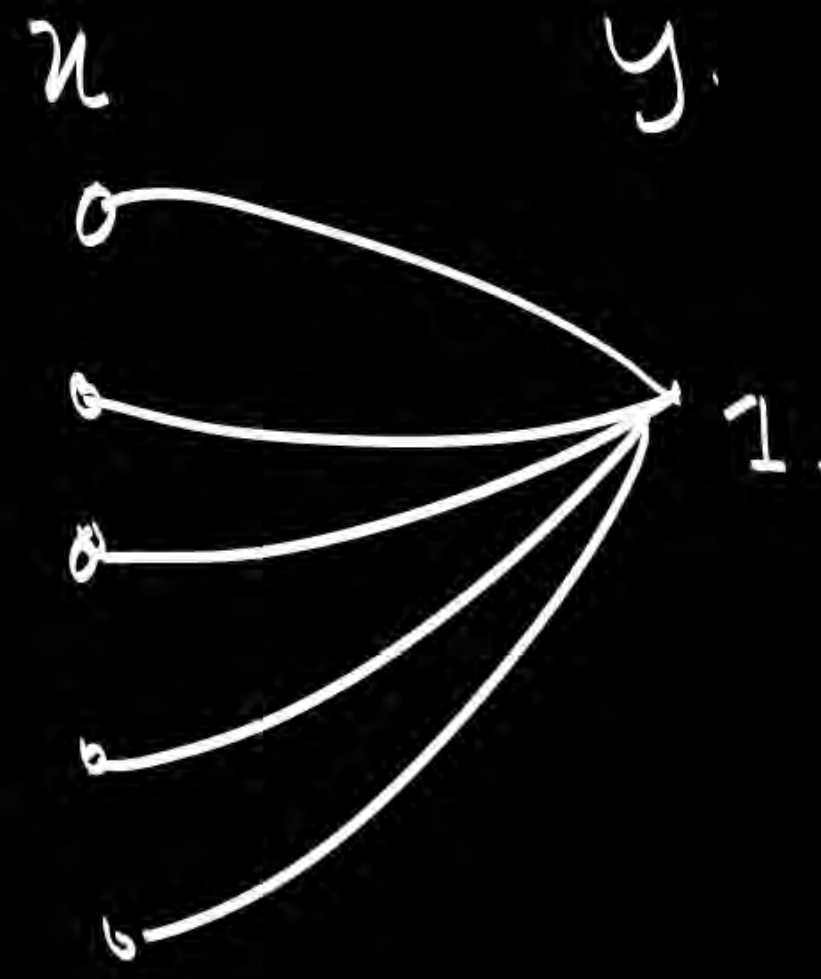
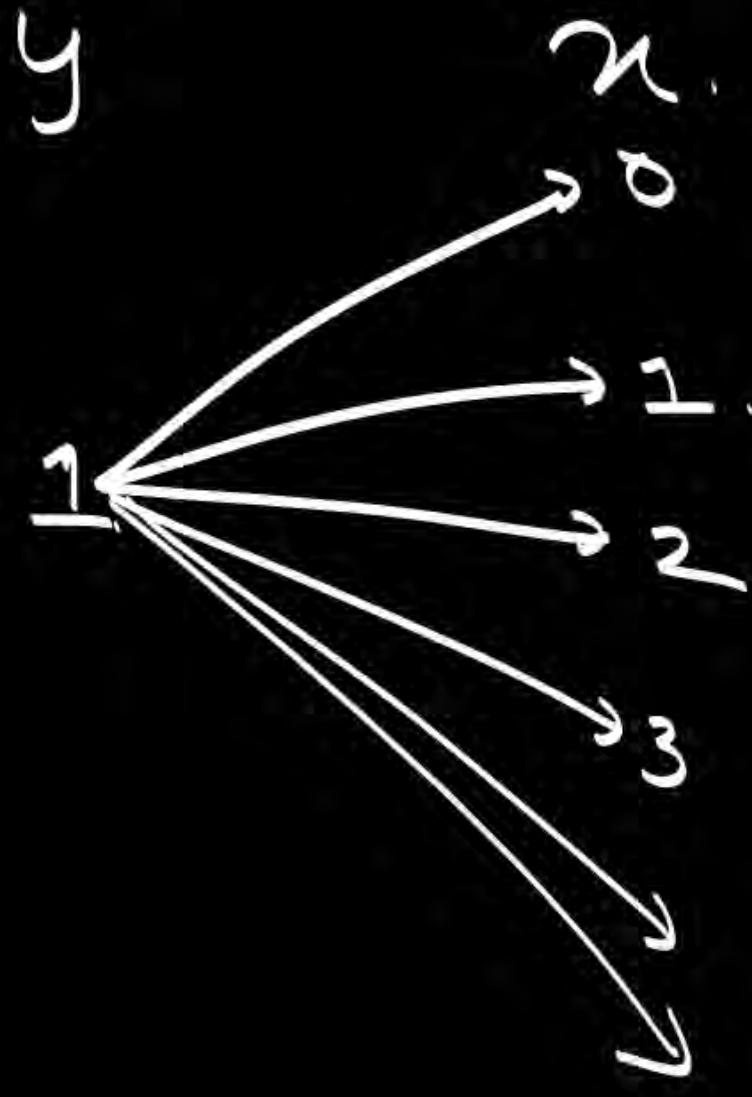


y demands to be
a single value or constant
which should point to
all values of x .



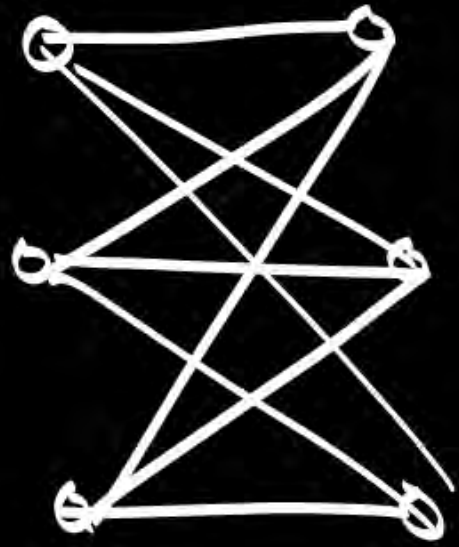
$$\underbrace{\forall x \exists y (x + y = 10)}_{\text{True}} \not\rightarrow \underbrace{\exists y \forall x (x + y = 10)}_{\text{false.}}$$

$$\exists y \forall x \left(\frac{x}{y} \in \mathbb{Z} \right) \longrightarrow \underbrace{\forall x}_{\text{true}} \underbrace{\exists y}_{\text{true}} \left(\frac{x}{y} \in \mathbb{Z} \right)$$



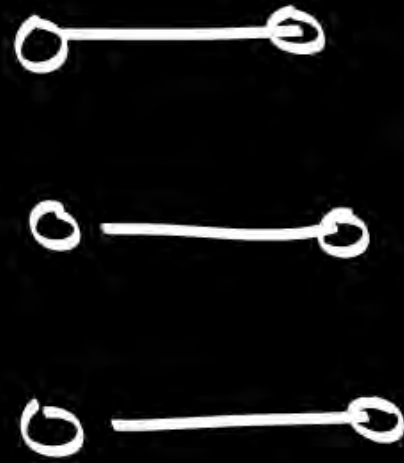
1.

$\forall x \forall y$

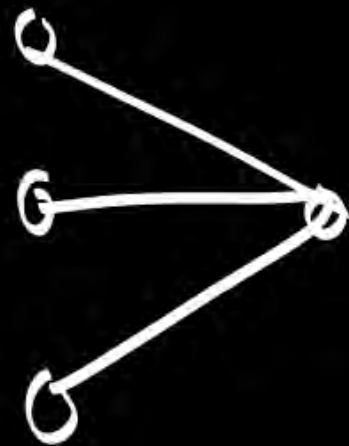


2.

$\forall x \exists y$

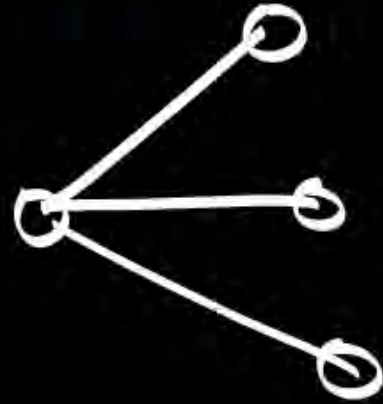


OR



3.

$\exists y \forall x$



4.

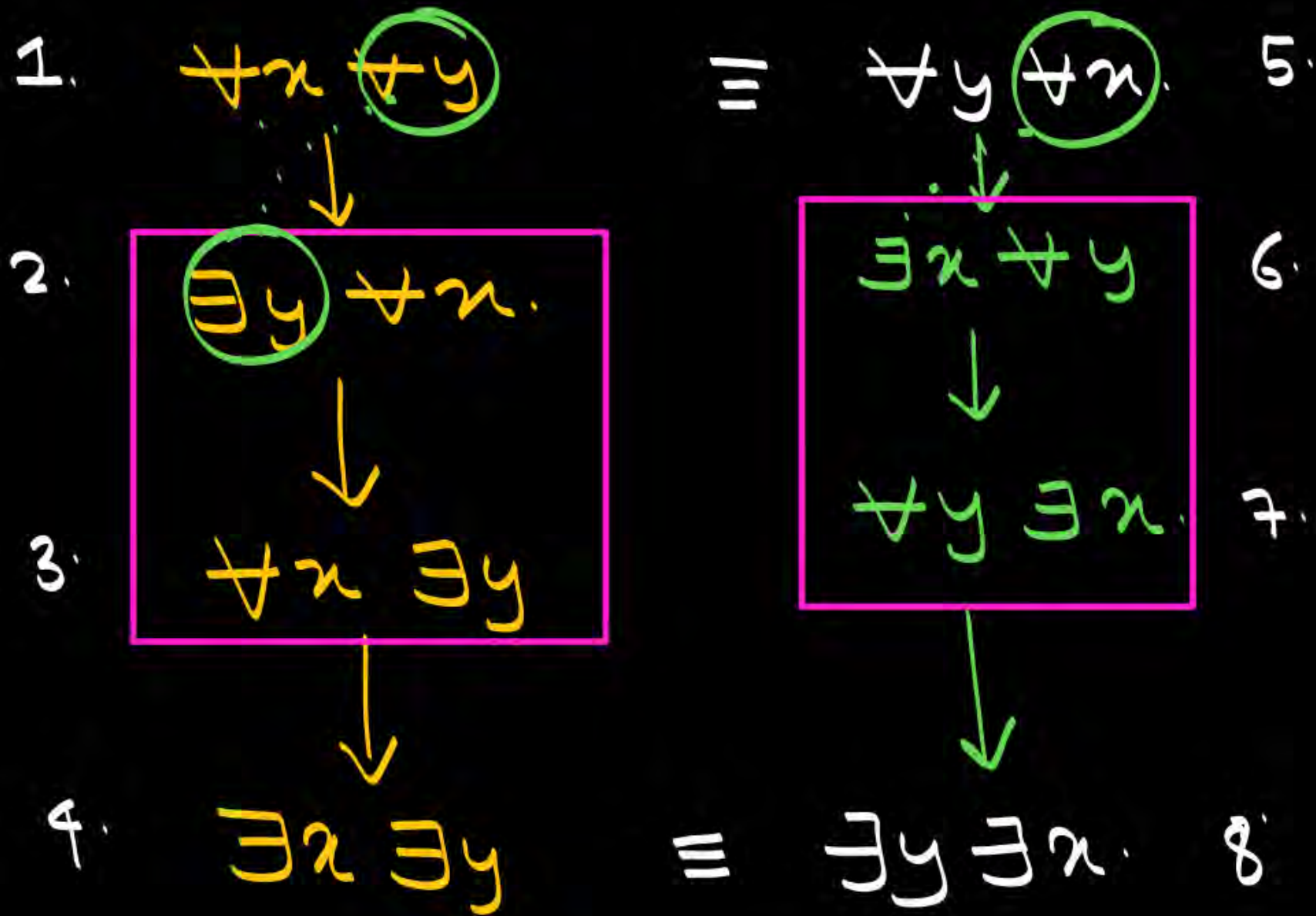
$\exists x \exists y$



1 $\rightarrow$ all

all $\rightarrow$ 4

3 $\rightarrow$ 2



1 $\rightarrow$ all
 5 $\rightarrow$ all.
 all $\rightarrow$ 4.
 all $\rightarrow$ 8.
 2, 3 & 6, 7
 no relation.



2 mins Summary

Topic

One

Logic Part 08

Topic

Two

Topic

Three

Topic

Four

Topic

Five



THANK - YOU